Please stick the barcode label here.

(b) Given that during night time the radiators maintain an average output power of 4.5 kW, how long can the radiators maintain this average power until the water temperature in the system drops to 60 °C ? Give your answer in hours. (2 marks)

(c) The rate of heat released by the solar water heating system during the time period calculated in (b) is in fact not constant and gradually drops. Explain why this is so. (1 mark)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

*2.

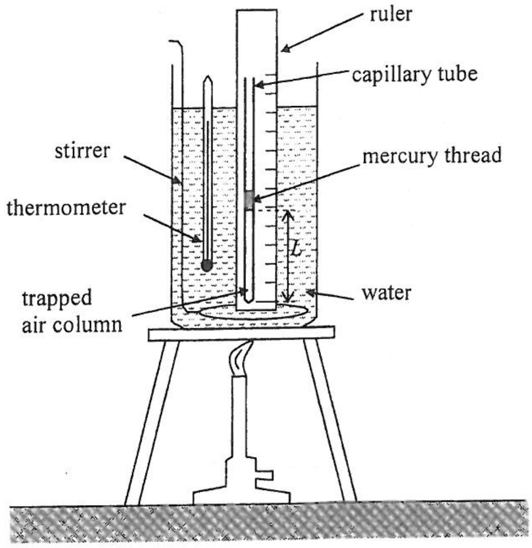

Figure 2.1

Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ. Some of the results are tabulated below:

Temperature θ / °C	20		92
Length of air column L / mm	64		80

(a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

Answers written in the margins will not be marked.

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Answers written in the margins will not be marked.

2022-DSE
PHY
PAPER 1B

B

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Candidate Number									

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2022

PHYSICS PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Graph paper and supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Question No.	Marks
1	8
2	10
3	12
4	6
5	8
6	9
7	9
8	10
9	6
10	6

* A 1 5 0 E 0 1 B *

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. A student conducts an experiment to study how to best retain the warmth of water by using identical paper cups X, Y and Z shown in Figure 1.1. Each cup is filled with 250 cm^3 of hot water and cup X serves as a control.

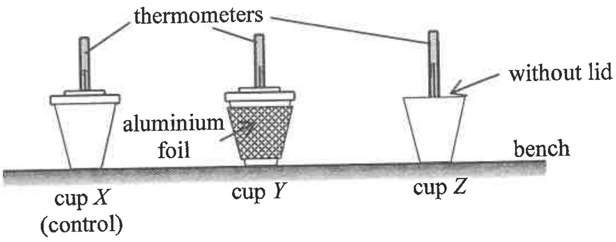

Figure 1.1

cup	wrapping	with a lid
X	no	yes
Y	aluminium wrapping	yes
Z	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (T) of water in the cups with time (t) elapsed.

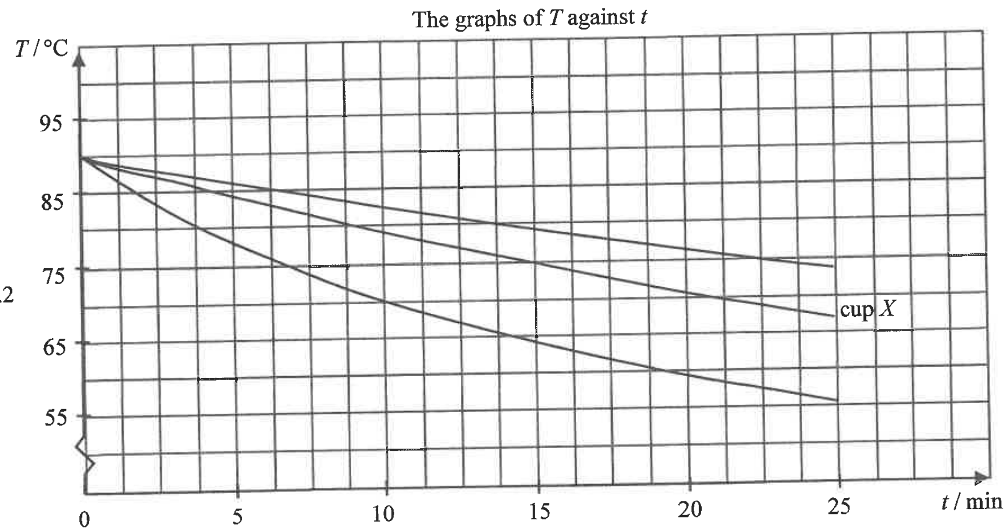

Figure 1.2

(a) Suggest a reason why this experiment begins with the **same initial water temperature** (90 °C). (1 mark)

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Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.